🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.6

Explore Integers and Their Opposites

Lesson 7-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.

Language Objective: I can explain my reasoning using the words integer, opposite, negative, positive, and number line.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Integer | **Opposite** | **Negative number** | **Positive number** | **Number line**

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A counting number, its opposite, or 0 is an ____________.
2. A number the same distance from 0 but on the other side is the ____________.
3. A number less than 0, written with a – sign, is a ____________.
4. A number greater than 0 is a ____________.
5. A line with numbers placed in order is a ____________.

Write About the Math

Write each play as an integer, then decide: did the team make a first down, and how far are they from the line of scrimmage?

Escribe cada jugada como un entero y luego decide: ¿logró el equipo un primer intento, y a qué distancia están de la línea de golpeo?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Integer** *Número entero*
- [ ] **Opposite** *Opuesto*
- [ ] **Negative number** *Número negativo*
- [ ] **Positive number** *Número positivo*
- [ ] **Number line** *Recta numérica*

Model: *In this situation 0 means ____ because ____.* — Student explains 0 is a chosen reference point (sea level), so values above it are positive and values below it are negative.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The 10-yard loss is the integer ____ and the 3-yard loss is the integer ____. Combining all four plays leaves the team ____ yards from where they started, which is ____ the line of scrimmage. Because they needed at least 10 yards, the team ____ make a first down.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Integer
2. **Notes 2:** Opposite
3. **Notes 3:** Negative number
4. **Notes 4:** Positive number
5. **Notes 5:** Number line
6. **Watch for this mistake:** *The most common mistake in this lesson is thinking that the opposite of a negative integer is still negative — for example, saying the opposite of -7 is -7, or that the opposite of -12 is -24. Opposites are the same DISTANCE from 0 but on OPPOSITE sides of 0, so taking an opposite always changes the direction and never changes the distance: the opposite of -7 is 7 and the opposite of -12 is 12. The one exception is 0, which is 0 units from itself and is therefore its own opposite. Before you submit, count the units from 0 and check that your answer is on the other side.*
7. **Try It 1:** A. 12°C and -12°C — *The opposite of -12 is 12, so 2 p.m. was 12°C. The opposite of 12 is -12, so 11 p.m. was back to -12°C.*
8. **Try It 2:** A. 12 yards past the line of scrimmage — *Write each play as an integer: -10, 4, -3, 21. Combining them gives -10 + 4 - 3 + 21 = 12, so the team ends 12 yards past the line of scrimmage.*